

Supplemental Appendix for “Should I State or Should I Show? Aligning AI with Human Preferences”

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This Supplemental Appendix provides additional details on the experimental design and supplementary analyses. Section 1 lists all lottery pairs used in Parts I and II. Section 2 reproduces the system prompts for Prompt-AI and Data-AI. Section 3 describes the estimation and predictive performance of the expected utility and cumulative prospect theory models. Section 4 presents word clouds of human-written and AI-generated prompts, describes the measures collected in Part III, and reports additional regressions incorporating these controls. Section 5 reports robustness analyses using GPT-5.4. Finally, Section 6 provides screenshots of the experimental instructions, tasks, and surveys.

1 Lottery Pairs

Table 1.1 summarizes all lottery questions and their corresponding parameters used in the experiment. Panel A shows all 13 lottery pairs used in Part I of the experiment, while Panel B shows all 13 lottery pairs used in Part II. Within each part, there are 2 easy questions, 5 hard questions, and 6 behavioral questions.

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TABLE 1.1: Lottery Pairs by Category: Part I and Part II

Panel A: Part I	Lottery A	Lottery B	$\mathbb{E}[B] - \mathbb{E}[A]$
<i>Easy</i>			
FOSD	(\$3.00, 0.60; \$1.00, 0.40)	(\$4.00, 0.70; \$2.00, 0.30)	+1.20
Easy	(\$1.50, 0.50; \$1.15, 0.50)	(\$1.55, 0.25; \$0.25, 0.75)	-0.75
<i>Hard</i>			
Hard 1	(\$3.85, 0.25; \$1.90, 0.75)	(\$4.70, 0.50; \$0.80, 0.50)	+0.36
Hard 2	(\$4.50, 0.50; \$0.50, 0.50)	(\$2.80, 0.25; \$2.25, 0.50;	-0.28
		\$1.60, 0.25)	
Hard 3	(\$10.00, 0.25; \$5.25, 0.25;	(\$6.75, 0.25; \$5.85, 0.25;	-0.36
		\$4.20, 0.25; \$0.30, 0.25)	
Hard 4	(\$4.05, 0.25; \$2.55, 0.25;	(\$4.30, 0.25; \$1.90, 0.25;	0.00
		\$1.50, 0.25; \$0.65, 0.25)	
SOSD	(\$2.00, 1)	(\$4.00, 0.50)	0.00
<i>Behavioral</i>			
AB ($p=0.8, r=0.25$)	(\$4.00, 0.80)	(\$3.00, 1)	-0.20
AB' ($p=0.8, r=0.25$)	(\$4.00, 0.20; \$3.00, 0.75)	(\$3.00, 1)	-0.05
CD ($p=0.8, r=0.25$)	(\$4.00, 0.20)	(\$3.00, 0.25)	-0.05
AB ($p=0.5, r=0.5$)	(\$5.00, 0.50)	(\$2.00, 1)	-0.50
AB' ($p=0.5, r=0.5$)	(\$5.00, 0.25; \$2.00, 0.50)	(\$2.00, 1)	-0.25
CD ($p=0.5, r=0.5$)	(\$5.00, 0.25)	(\$2.00, 0.50)	-0.25
Panel B: Part II	Lottery A	Lottery B	$\mathbb{E}[B] - \mathbb{E}[A]$
<i>Easy</i>			
FOSD	(\$2.00, 0.30; \$1.00, 0.50)	(\$3.00, 0.50; \$2.00, 0.20;	+1.10
Easy	(\$15.00, 0.50; \$11.50, 0.50)	\$1.00, 0.30)	-7.50
<i>Hard</i>			
Hard 1	(\$38.50, 0.25; \$19.00, 0.75)	(\$47.00, 0.50; \$8.00, 0.50)	+3.63
Hard 2	(\$5.00, 0.50)	(\$3.00, 0.25; \$2.75, 0.50;	-0.13
		\$1.00, 0.25)	
Hard 3	(\$9.25, 0.25; \$6.00, 0.25;	(\$7.30, 0.25; \$5.50, 0.25;	-0.36
		\$3.50, 0.25; \$1.00, 0.25)	
Hard 4	(\$3.75, 0.25; \$3.00, 0.25;	(\$4.50, 0.25; \$2.50, 0.25;	0.00
		\$2.00, 0.25)	
SOSD	(\$3.00, 0.25; \$2.00, 0.50;	\$1.25, 0.25; \$0.50, 0.25)	0.00
<i>Behavioral</i>			
AB ($p=10/11, r=0.11$)	(\$3.00, 0.95)	(\$4.00, 0.91)	+0.79
AB' ($p=10/11, r=0.11$)	(\$3.00, 0.95)	(\$4.00, 0.10; \$3.00, 0.89)	+0.22
CD ($p=10/11, r=0.11$)	(\$3.00, 0.11)	(\$4.00, 0.10)	+0.07
AB ($p=0.3, r=0.5$)	(\$1.00, 0.95)	(\$5.00, 0.30)	+0.55
AB' ($p=0.3, r=0.5$)	(\$1.00, 0.95)	(\$5.00, 0.15; \$1.00, 0.50)	+0.30
CD ($p=0.3, r=0.5$)	(\$1.00, 0.50)	(\$5.00, 0.15)	+0.25

Note: Lotteries are represented as $(x_1, p_1; \dots, x_n, p_n)$, where each outcome x_i occurs with probability $p_i \geq 0$ and $\sum_{i=1}^n p_i = 1$. To simplify notation, we omit the zero outcome and use (x, p) to denote the prospect $(x, p; 0, 1 - p)$.

2 AI System Prompts

This section reproduces the system prompts used for the Prompt-AI and Data-AI agents verbatim. Both AI agents additionally receive the 13 Part II lottery pairs on which they need to make decisions on behalf of human subjects as a user message.

Prompt-AI System Prompt

You are an AI assistant helping a participant in an economics experiment make decisions between lottery pairs.

Based on the participant's written instructions shown below, infer their preferences and make the choices they would make for new lottery pairs.

You will now be presented with lottery choices. After reasoning through each choice, you MUST end your response with a JSON object containing your choices in this exact format:

```
{"post_1": "A", "post_2": "B", ..., "post_13": "A"}
```

Replace A/B with your actual choices. The JSON must be the last thing in your response.

PARTICIPANT'S WRITTEN INSTRUCTIONS:

"""

[PARTICIPANT'S WRITTEN PROMPT -- inserted verbatim]

"""

Data-AI System Prompt

You are an AI assistant helping a participant in an economics experiment make decisions between lottery pairs.

Based on the participant's previous choices shown below, infer their preferences and make similar choices for new lottery pairs.

You will now be presented with lottery choices. After reasoning through each choice, you MUST end your response with a JSON object containing your choices in this exact format:

```
{"post_1": "A", "post_2": "B", ..., "post_13": "A"}
```

Replace A/B with your actual choices. The JSON must be the last thing in your response.

PARTICIPANT'S PREVIOUS CHOICES (13 questions):

- Q1: Chose Lottery [X] (A: [description], B: [description])
- Q2: Chose Lottery [X] (A: [description], B: [description])
- ...

[All 13 Part I choices with full lottery descriptions]

3 Structural Exercise

This section describes our structural exercises in detail. We estimate two structural models of risky choice from subjects' Part I choices—an expected utility (EU) model and a cumulative prospect theory (CPT) model—and use each model to predict subjects' choices in Part II.

Expected Utility. For each subject, we estimate a constant relative risk aversion (CRRA) expected utility model with a logit choice error using the 13 lottery choices in Part I. Given a lottery $L = (x_i, p_i)_{i=1}^n$, the expected utility is

$$EU(L; \rho) = \sum_{i=1}^n p_i u(x_i; \rho), \text{ where } u(x; \rho) = \begin{cases} \frac{(x+w)^{1-\rho}}{1-\rho} & \text{if } \rho \neq 1, \\ \ln(x+w) & \text{if } \rho = 1, \end{cases}$$

ρ is the coefficient of relative risk aversion, and $w = 0.01$ is a small wealth offset that ensures the utility function is defined at zero payoffs. For a lottery pair (A, B) , the probability of lottery A being chosen by the agent with CRRA parameter ρ is thus

$$\mathbb{P}(A; \rho, \mu) = \frac{1}{1 + e^{-\mu[EU(A; \rho) - EU(B; \rho)]}},$$

where $\mu \geq 0$ is a precision parameter. As $\mu \rightarrow \infty$, the agent deterministically chooses the higher-EU lottery. As $\mu \rightarrow 0$, their choices approach uniform random choice.

For each subject, we observe 13 choices in Part I and use maximum likelihood to estimate their individual (ρ, μ) . The log-likelihood is

$$\ell(\rho, \mu) = \sum_{k=1}^{13} [\mathbb{1}_{c_k=A_k} \ln \mathbb{P}(A_k; \rho, \mu) + \mathbb{1}_{c_k=B_k} \ln \mathbb{P}(B_k; \rho, \mu)],$$

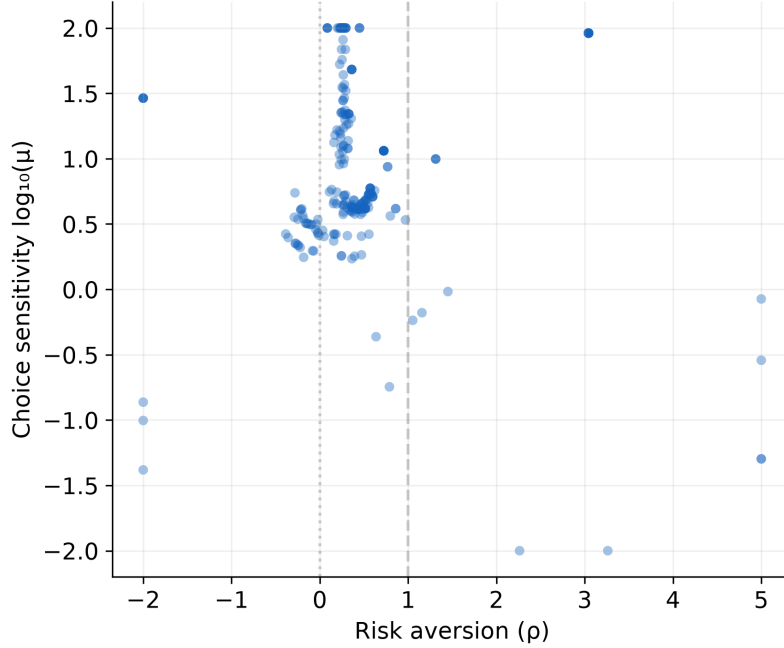
where $c_k \in \{A_k, B_k\}$ is the observed choice from binary lottery menu $k = 1, \dots, 13$. We then optimize over the bounded parameter space $\rho \in [-2, 5]$, $\mu \in [0.01, 100]$ using L-BFGS-B, with a grid of 24 starting points ($\rho_0 \in \{-0.5, 0, 0.5, 1, 2, 3\}$, $\mu_0 \in \{0.1, 1, 5, 20\}$) to mitigate local optima. Finally, we retain the solution with the largest log-likelihood.

Figure 3.1 shows a scatter plot of our estimated parameters for each subject. Most subjects are moderately risk averse: 70% have estimated CRRA parameters ρ between zero and one. Among these subjects, the median ρ is 0.34, and the median precision parameter is 5.9.

We then use these estimated risk parameters to predict subjects' choices in Part II. For each lottery pair, we predict the lottery with the higher estimated expected utility. We refer to these as predictions from the expected utility (EU) model. As described in the main text, the EU model achieves an average match rate of about 75%, similar to that of Data-AI. Moreover, at the subject-question level, EU predictions are more closely aligned with Data-AI than with Prompt-AI, matching the former 80% of the time and the latter 73% of the time. Across question categories, we also find that the EU model is less predictive for hard and behavioral questions: the average match rates for easy, behavioral, and hard are 99%, 72%, and 70%, respectively.

Cumulative Prospect Theory. We next estimate a cumulative prospect theory model with two preference parameters and a logit precision parameter (Tversky and

FIGURE 3.1: Scatter plot of estimated $\hat{\mu}$ and $\hat{\rho}$ values per subject, based on their Part I choices.



Note: Dashed lines at $\rho = 0$ and $\rho = 1$ indicate risk neutrality and log utility, respectively.

Kahneman, 1992). The Bernoulli utility function is the same CRRA utility $u(x; \rho)$ as above, and probabilities are distorted by the weighting function

$$w(p) = \frac{p^\gamma}{p^\gamma + (1 - p)^\gamma},$$

where $\gamma \in [0, 1]$ is the distortion parameter (Goldstein and Einhorn, 1987). This corresponds to the most complete one-parameter weighting function from Fudenberg et al. (2026). The parameter $\gamma = 1$ corresponds to no probability distortion, while lower values imply stronger inverse S-shaped probability weighting.

We order all non-negative lottery outcomes so that $x_1 \geq x_2 \geq \dots \geq x_n$. For a given γ , the CPT value of a lottery $L = (x_i, p_i)_{i=1}^n$ is

$$CPT(L; \rho, \gamma) = \sum_{i=1}^n \pi_i u(x_i; \rho), \quad \text{where } \pi_i = w\left(\sum_{j \leq i} p_j\right) - w\left(\sum_{j < i} p_j\right).$$

We again estimate this model with a logit choice error as above with precision pa-

parameter μ , so that $\gamma = 1$ nests the above expected utility functional form. For each subject, we use their 13 choices in Part I to estimate (ρ, γ, μ) by maximum likelihood over the bounded parameter space $\rho \in [-2, 5]$, $\gamma \in [0.05, 1]$, $\mu \in [0.01, 100]$. We impose the restriction $\gamma \leq 1$ which yields the standard inverse-S shape. We again use L-BFGS-B to estimate, with a grid of 72 starting points ($\rho_0 \in \{-0.5, 0, 0.5, 1, 2, 3\}$, $\gamma_0 \in \{0.3, 0.6, 1\}$, $\mu_0 \in \{0.1, 1, 5, 20\}$) to mitigate local optima. As before, we retain the solution with the largest log-likelihood.

The estimates indicate moderate risk aversion and probability weighting. The median estimates of $(\hat{\rho}, \hat{\gamma}, \hat{\mu})$ are (0.30, 0.84, 12.92), with 35% of subjects having $\hat{\gamma}$ at the upper bound of one, i.e., no probability distortion.

We then use the estimated parameters $(\hat{\rho}, \hat{\gamma})$ to predict subjects’ choices in Part II, selecting the lottery with the higher estimated CPT value. Overall, the CPT model performs nearly identically to the EU model. It achieves an average match rate of 75%, which is statistically indistinguishable from both the EU model (paired t -test: $t = 0.67$, $p = 0.502$) and Data-AI (paired t -test: $t = 1.35$, $p = 0.177$). At the subject-question level, CPT predictions coincide with EU predictions 92% of the time, and align with Data-AI predictions 78% of the time, compared with 71% for Prompt-AI. Across question categories, the CPT model is also less predictive for hard and behavioral questions: the average match rates for easy, behavioral, and hard questions are 99%, 72%, and 70%, respectively.

4 Additional Analysis

In this section, we first describe the additional controls measured in Part III of the experiment. We also present the word clouds comparing human-written and AI-generated prompts (Figure 4.1). We then report three additional regression tables. Table 4.1 regresses the subject-level difference between Data-AI and Prompt-AI match rates on these controls. Table 4.2 reports the full regression results corresponding to Column (4) of Table 2 in the main text. Finally, Table 4.3 extends Table 1 in the main text by adding the number of behavioral effects and the same set of controls.

Additional controls in Part III

- *AI Comfort* and *Writing Comfort* are single 1–7 Likert items elicited at the end of the experiment: “How comfortable are you with using AI tools (e.g.,

ChatGPT, Claude, Copilot)?” and “How comfortable are you with writing instructions or explanations for others?”, respectively, with endpoints labeled “Not at all comfortable” and “Very comfortable”.

- *Impatience* is a single 1–7 Likert item: “How would you rate your level of impatience in general?”, with endpoints labeled “Very patient” and “Very impatient”.
- *Education Level*, *AI Usage Frequency*, and *Management Experience* are ordered categorical survey items, entered in the regressions as ordinal scores. *Education Level* (“What is the highest level of education you have completed?”) ranges from high school or equivalent to a doctorate or professional degree (six categories, coded 0–5). *AI Usage Frequency* (“How often do you use AI tools (e.g., ChatGPT, Claude, Copilot) in your daily life or work?”) ranges from never to daily (five categories, coded 0–4). *Management Experience* (“Do you have experience managing or supervising other people?”) ranges from no management experience to extensive experience of six or more years (four categories, coded 0–3).
- *IQ (Raven’s)* is the number of correct answers on a 6-item matrix reasoning test from the International Cognitive Ability Resource (ICAR; [Condon and Revelle, 2014](#)) (score 0–6). Subjects receive 20¢ for each correct answer.
- *Measures of Overconfidence* consist of two types. The first is the difference between the subject’s self-reported guess of their own score and their actual IQ test score, which captures absolute overconfidence. The second is the difference between the subject’s self-reported guess of their own score and their performance relative to other Prolific subjects, which captures relative overconfidence. Subjects receive 25¢ for each correct answer.
- *Risky Investment* is a variable that records how many tokens (out of 100 available) a subject chose to invest in a risky project that pays back 2.5 times the invested amount with probability 50%, and zero otherwise. *Compound Investment* is a variable that records how many tokens (out of the 100 available) a subject chose to invest in a risky project that has the same odds as the one described in the Risky Investment except that the success of the investment was determined by a compound lottery that reduces to a 50% chance of success.

We report *Compound Investment* relative to *Risky Investment*, calculated as the amount invested in the compound lottery minus the amount invested in the simple lottery. A negative difference indicates compound-lottery aversion, while a positive difference indicates a preference for compound lotteries.

- The “Big Five” personality traits (*Extraversion*, *Agreeableness*, *Conscientiousness*, *Emotional Stability*, *Openness*) are measured using the Ten-Item Personality Inventory (TIPI; Gosling et al., 2003), with each trait based on two 1–7 Likert items. There are two questions for each trait, one of which is reverse-coded. For example, *Extraversion* is measured as the average of the response to “I see myself as extraverted, enthusiastic” and the reverse-coded response to “I see myself as reserved, quiet”.

FIGURE 4.1: Word cloud of human-written and AI-generated prompts



(a) Word cloud of human-written prompts.



(b) Word cloud of AI-generated prompts.

Note: This figure presents word clouds of the top 100 words that are most likely to appear in each set of prompts, with the size of each word representing its relative frequency.

TABLE 4.1: Relative Match Rates and Part III Controls

	Match Rate Difference (Data-AI – Prompt-AI)
AI Comfort	-0.025* (0.013)
Writing Comfort	0.010 (0.011)
Impatience	0.023** (0.010)
IQ (Ravens)	0.000 (0.012)
Risky Investment	-0.014 (0.009)
Compound Investment	-0.024** (0.010)
Overconfidence (abs.)	-0.009 (0.016)
Overconfidence (rel.)	0.009 (0.015)
Extraversion	-0.010 (0.012)
Agreeableness	0.006 (0.012)
Conscientiousness	-0.012 (0.012)
Emot. Stability	0.016 (0.014)
Openness	-0.013 (0.012)
Education Level	0.002 (0.011)
AI Usage Frequency	0.007 (0.012)
Management Experience	0.008 (0.010)
Age	-0.014 (0.010)
Female	0.003 (0.010)
Constant	0.034*** (0.009)
Observations	266
Subjects	266
R^2	0.092

Note: Robust standard errors are reported in parentheses. Compound investment is the amount invested in the compound lottery minus the amount invested in its simple counterpart. Negative values indicate compound-lottery aversion, while positive values indicate a preference for compound lotteries. All reported controls are standardized using the subject-level sample mean and sample standard deviation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE 4.2: Match Rates and Behavioral Effects

	Match rate
Data-AI (vs Prompt-AI)	0.010 (0.013)
No. Behavioral Effects	-0.042*** (0.008)
Data-AI $\times$ No. Behavioral Effects	0.015** (0.007)
AI Comfort	-0.007 (0.010)
Writing Comfort	0.003 (0.009)
Impatience	0.017** (0.008)
IQ (Ravens)	0.022** (0.011)
Risky Investment	-0.001 (0.007)
Compound Investment	-0.004 (0.008)
Overconfidence (abs.)	-0.010 (0.010)
Overconfidence (rel.)	0.012 (0.010)
Extraversion	-0.012 (0.008)
Agreeableness	0.008 (0.008)
Conscientiousness	0.000 (0.009)
Emot. Stability	0.004 (0.010)
Openness	-0.002 (0.008)
Education Level	0.003 (0.008)
AI Usage Frequency	-0.004 (0.009)
Management Experience	0.009 (0.008)
Age	0.007 (0.009)
Female	-0.005 (0.008)
Constant	0.773 (0.014)
Observations	532
Subjects	266
R^2	0.153

Note: Standard errors clustered by subject are reported in parentheses. Data-AI is an indicator variable that takes value one if the AI agent was given the subjects' choices instead of their written prompt, and zero otherwise. No. Behavioral effects is a count variable ranging from 0 to 4. Compound investment is the amount invested in the compound lottery minus the amount invested in its simple counterpart. All regressors, except no. behavioral effects, the indicator for Data-AI, and their interaction, are standardized using the subject-level sample mean and sample standard deviation.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE 4.3: Textual Similarity, AI Agreement, and Match Rate

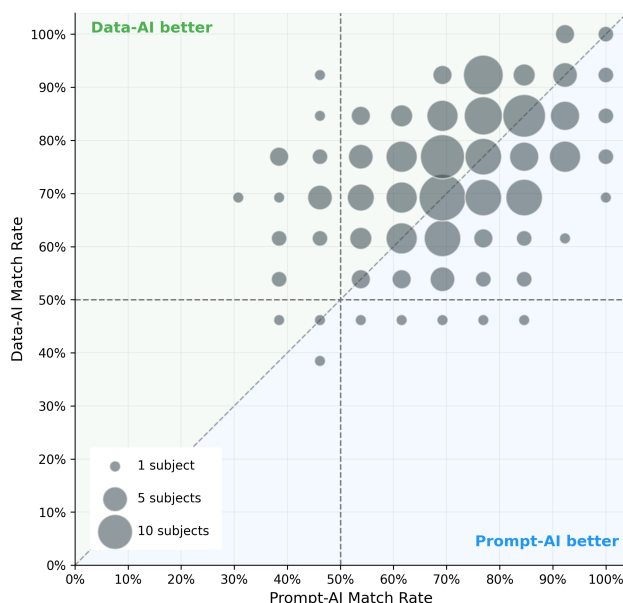
	(1)	(2)
	Agreement Rate	Absolute Match Rate Gap
	AI-generated vs. Human-written prompt	Data-AI vs. Prompt-AI
SBERT Cosine Similarity	0.228*** (0.088)	-0.081* (0.049)
No. Behavioral Effects	-0.033*** (0.010)	0.011** (0.006)
AI Comfort	-0.020 (0.018)	-0.004 (0.008)
Writing Comfort	-0.004 (0.014)	0.011 (0.008)
Impatience	0.017 (0.015)	0.001 (0.008)
IQ (Ravens)	-0.022 (0.017)	-0.012 (0.009)
Risky Investment	-0.023* (0.012)	0.006 (0.007)
Compound Investment	-0.006 (0.012)	-0.013* (0.008)
Overconfidence (abs.)	-0.008 (0.019)	-0.006 (0.010)
Overconfidence (rel.)	-0.026 (0.019)	0.001 (0.010)
Extraversion	0.007 (0.015)	-0.008 (0.009)
Agreeableness	-0.010 (0.017)	-0.004 (0.008)
Conscientiousness	-0.001 (0.017)	-0.001 (0.009)
Emot. Stability	0.027 (0.020)	0.005 (0.010)
Openness	-0.037** (0.015)	0.000 (0.009)
Education Level	-0.019 (0.013)	0.001 (0.007)
AI Usage Frequency	0.003 (0.017)	-0.004 (0.008)
Management Experience	0.007 (0.014)	-0.002 (0.007)
Age	-0.003 (0.014)	-0.009 (0.007)
Female	-0.007 (0.012)	-0.004 (0.007)
Constant	0.649 (0.064)	0.142 (0.036)
Observations	266	266
R^2	0.155	0.095

Note: Both dependent variables are measured in Part II. Robust standard errors in parentheses. No. Behavioral effects is a count ranging from 0 to 4. Compound investment is the amount invested in the compound lottery minus the amount invested in its simple counterpart. All regressors, except SBERT cosine similarity and behavioral effects, are standardized using the subject-level sample mean and sample standard deviation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 GPT-5.4 Analysis

This section replicates the main analysis using GPT-5.4 (gpt-5.4) as the underlying AI model instead of Claude Opus 4.5. Tables 5.1 and 5.2 report the GPT-5.4 counterparts of main-text Tables 1 and 2, respectively. Figures 5.1 and 5.2 reproduce the analysis in main-text Figures 1 and 2, respectively.

FIGURE 5.1: Comparison of Prompt-AI and Data-AI match rates (GPT-5.4).



Note: Each bubble plots a subject’s Data-AI match rate against their Prompt-AI match rate, with bubble area proportional to the number of subjects at that point. Subjects above the 45-degree line are matched better by Data-AI; subjects below it are matched better by Prompt-AI. The dashed grey lines at 50% mark the expected match rate of random guessing on each axis.

TABLE 5.1: Textual Similarity, AI Agreement, and Match Rate (GPT-5.4)

	(1)	(2)
	Agreement Rate	Absolute Match Rate Gap
	AI-generated vs. Human-written prompt	Data-AI vs. Prompt-AI
SBERT Cosine Similarity	0.200*** (0.074)	-0.089** (0.044)
Constant	0.619 (0.048)	0.166 (0.028)
Observations	267	267
R^2	0.026	0.016

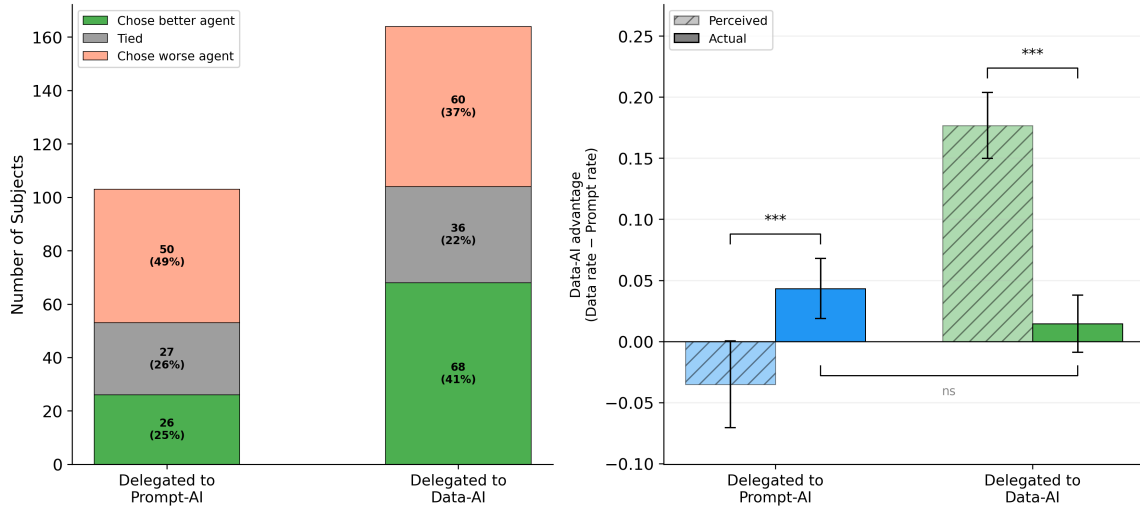
Note: Both dependent variables are measured in Part II. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE 5.2: Match Rates and Behavioral Effects (GPT-5.4)

	Match rate			
	(1)	(2)	(3)	(4)
Data-AI (vs Prompt-AI)	0.026*** (0.009)	0.026*** (0.009)	-0.021 (0.013)	-0.021 (0.014)
No. Behavioral Effects		-0.022*** (0.005)	-0.037*** (0.007)	-0.043*** (0.007)
Data-AI $\times$ No. Behavioral Effects			0.030*** (0.008)	0.030*** (0.008)
Constant	0.717 (0.009)	0.752 (0.013)	0.776 (0.014)	0.785 (0.014)
Controls	No	No	No	Yes
Observations	532	532	532	532
Subjects	266	266	266	266
R^2	0.009	0.046	0.063	0.136

Note: Standard errors clustered by subject in parentheses. Each subject contributes two match rate observations (Prompt-AI and Data-AI). Data-AI is an indicator variable for whether the match rate is for Data-AI (versus Prompt-AI). No. Behavioral Effects is a count variable ranging from 0 to 4. Column (4) adds controls collected in Part III. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

FIGURE 5.2: Delegation Choice: Realized Accuracy and Beliefs (GPT-5.4)



Note: *Left:* counts (within-group percentages) choosing the better agent (green), tied agents (grey), or the worse agent (salmon). *Right:* mean Data-AI-Prompt-AI match-rate differences, actual (solid) and perceived (hatched), with 95% confidence intervals. Upper brackets: paired t-tests of perceived versus actual differences within delegation groups. Lower bracket: two-sample t-test of actual differences across groups. Delegation decisions and perceived advantages are based on Claude Opus 4.5; realized accuracy and actual advantages use GPT-5.4 match rates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Experimental Instructions and Screenshots

6.1 Informed Consent

Informed Consent

Please carefully review the information below and click either "I have read the information above and I consent to participate in this study" or "I do not consent to participate". This study has been approved by the Human Research Ethics Committee at Monash University (Project ID: 50731).

The full Explanatory Statement is available for download [here](#). A summary of the key information is provided below.

Principal investigators

Dr Keaton Ellis (keaton.ellis@monash.edu) and Dr Kate Huang (kate.huang@monash.edu), Monash University.

What the study is about

This research project aims to learn more about decision-making under uncertainty and how people delegate decisions to artificial intelligence (AI) systems.

What we will ask you to do

If you choose to participate, we will ask you to make a series of choices under uncertainty and to decide whether to delegate your decisions to an AI system in similar situations. We will analyze participants' decisions in an anonymous manner (your identity will not be recorded in the dataset) to further our understanding of how people make decisions under uncertainty and when they choose to delegate decisions to an AI system.

Duration

The study will last about 25-30 minutes.

Compensation for participation

You will receive a participation fee of **\$5.00** for completing all parts of this study. Additionally, you will have a chance to earn additional bonus (up to **\$51.70**). How much you receive depends partly on your decisions and partly on chance. All earnings will be paid to you via Prolific.

Privacy/Confidentiality/Data Security

We will treat all your responses in this study anonymously and use your Prolific ID for payment purposes only. Once the study is complete, we will remove this information from our data and maintain the de-identified data in a password-protected folder accessible only by the research team members. We will only report results in de-identified form. Please note that the study is hosted on an external server not affiliated with Monash University and with its own privacy and security policies. We anticipate that your participation in this study presents no greater risk than your everyday use of the Internet.

Taking part is voluntary

Taking part in this study is voluntary, and you may withdraw from the study at any time. Withdrawing before the study is complete will mean that you will not earn a participation fee and will not be eligible for any additional compensation from this study.

Contact Information

If you have any questions or concerns regarding your rights as a subject of this study, you may contact the Monash University Human Research Ethics Committee at MUHREC@monash.edu

Do you consent to participate in this study?

- ☐ I have read the information above and I consent to participate in this study.
- ☐ I do not consent to participate.

Next

FIGURE 6.1: Informed Consent

6.2 Study Overview and Instructions

Study Overview

Thank you for participating. Please read these instructions carefully.

Study Structure

This study has **three parts**. You will receive detailed instructions for each part at its start.

- **Part I** has three tasks. You will provide multiple pieces of information to help AI agents make choices between lotteries on your behalf. We will explain later in detail what each task entails.
- **Part II** has one task. You will make choices between lotteries.
- **Part III** asks you to answer additional survey questions and complete a short reasoning test.

Payment

You will receive:

- **Base payment:** \$5.00 for completing the study
- **Lottery bonus:** We will randomly select a choice from all choices made during Parts I and II. The selected choice might have been made by you, or might have been made by an AI agent on your behalf. The selected choice (i.e. a lottery) in the question will then be played out for real money. The highest possible lottery outcome is \$47.00.
- **Survey bonus:** Some survey questions in Part III involve bonus payments. We will explain later in detail how the bonuses will be determined.

Please complete each task carefully, as choices from each task could be selected for your payment.

While answer the questions, we ask you to **not browse the internet** and **not consult with others**. We are genuinely interested in your opinions and preferences.

Next

FIGURE 6.2: Study Overview

Part I Instructions

You are now reading the instructions for **Part I**. In Part I, you will work with lotteries and help AI agents make lottery choices on your behalf.

Understanding Lotteries

A **lottery** is a gamble with uncertain outcomes. Each lottery shows you the possible amounts you could win and the probability (chance) of each outcome.

Throughout this study, you will choose between pairs of lotteries presented like this:

Lottery A

Probability	Outcome
50%	\$4.00
50%	\$0.00

Lottery B

Probability	Outcome
100%	\$1.50

Lottery A means there is a 50% chance you win \$4.00 and a 50% chance you win \$0.00.

Lottery B means you win \$1.50 for certain.

When you choose between two lotteries, you are selecting which gamble you prefer to play.

Next

FIGURE 6.3: Part I Instructions: Understanding Lotteries

Part I Instructions

Additional Information About Lotteries

When picking, you will see five additional pieces of information:

Average Payment: (also known as "expected value") This is the average payment the lottery would pay out if it were played many, many times. Lotteries with higher average payments pay more on average, but there is still randomness to how much they pay each time. To calculate the average payment, we multiply each outcome by the probability of that outcome occurring and add this up for all the outcomes.

Payment Variability: (also known as "variance") This is a measure of how much payments can vary. Lotteries with higher payment variability pay amounts that are more spread out, which often means the difference between the larger and lower payment is larger. To calculate the payment variability, we subtract each outcome from the average payment, square it, and then add this up for all the outcomes, weighted by the respective probabilities.

Minimum Payment: This is the minimum possible amount the lottery could pay.

Maximum Payment: This is the maximum possible amount the lottery could pay.

Chance of Max Payment: This is the probability that the lottery pays the Maximum Payment.

We will show you two lotteries, and you can access this additional information about them by clicking on the buttons at the bottom, as we show below. When you click on a button, it will display that information for both lotteries:

Lottery A		Lottery B	
Probability	Outcome	Probability	Outcome
50%	\$4.00	100%	\$1.50
50%	\$0.00		

Additional information:

Average Payment

Payment Variability

Minimum Payment

Maximum Payment

Chance of Max

Please click on the buttons above to get a sense of how this works for the example lotteries above.

When making your choices between lotteries, you will be able to click to learn as many pieces of additional information you want about the lotteries. If you wish, you can use this information to help you make your choices. **Please note that you do not have to use these buttons at all; they are just for your convenience.**

When you are ready, continue.

Next

FIGURE 6.4: Part I Instructions: Additional Information About Lotteries

Comprehension Quiz

Attempt 1 of 3

Please answer all questions correctly to continue. You have **3** attempt(s) remaining.

Question 1:

How will your bonus payment be determined?

- ☐ The outcomes of all lottery choices will be added together.
- ☐ One lottery choice will be randomly selected and played out.
- ☐ The lottery with the highest expected value will be played.

Question 2:

If a lottery has a 50% chance of \$4.00 and 50% chance of \$0.00, what are the possible outcomes?

- ☐ You could win either \$4.00 or \$0.00
- ☐ You will always win \$2.00 (the average)
- ☐ You will always win \$4.00

Question 3:

When writing instructions for the AI, what should you aim for?

- ☐ Write instructions that make the AI choose randomly.
- ☐ Write instructions that lead the AI to make choices you prefer.
- ☐ Write instructions that make the AI respond as quickly as possible.

Question 4:

What is the maximum total bonus you can earn?

- ☐ \$1.00
- ☐ \$10.00
- ☐ \$51.70

Next

FIGURE 6.5: Comprehension Quiz

6.3 Part I, Task 1: Initial Lottery Choices

Part I, Task 1: Initial Lottery Choices

In this task, you will make **13 choices** between pairs of lotteries. In each choice, you will see two lotteries with different probabilities and potential payoffs. Please select the lottery you prefer to play.

Your choices will be shown to an AI agent (Claude, developed by Anthropic). Based on your **13 choices**, the AI agent will then make choices on your behalf between **new pairs of lotteries that are similar in structure but not identical** to those in this task. Because one of the AI agent's choices may be used to determine your bonus payment, it is important that you choose the lotteries you truly prefer so that the AI agent can learn your preferences accurately.

Next

FIGURE 6.6: Part I, Task 1 Introduction

Question 1 of 13

Click on the lottery you prefer:

Lottery A

Probability	Outcome
20%	\$4.00
80%	\$0.00

Lottery B

Probability	Outcome
25%	\$3.00
75%	\$0.00

Click to compare statistics:

Average Payment

Payment Variability

Minimum Payment

Maximum Payment

Chance of Max

Next

FIGURE 6.7: Lottery Choice Screen (Part I)

Can you spot the animal camouflaged in the image below?

Please click on the photo where you think the animal is located

This task does not affect your payment and is just for fun.



This photo was originally published by the Reader's Digest on August 22, 2019 (<https://www.rd.com/list/camouflaged-animal-photos/>)

Next

FIGURE 6.8: Attention Break

6.4 Part I, Task 2: Writing Instructions for an AI

Part I, Task 2: Writing Instructions for an AI

In this task, you will **write instructions** to an AI agent (Claude, developed by Anthropic). Based on your instructions, the AI agent will then make choices on your behalf between **new pairs of lotteries that are similar in structure but not identical** to those shown in Task 1. Because one of the AI agent's choices may be used to determine your bonus payment, it is important that you write the instructions carefully so that the AI agent can learn your preferences accurately.

Note: Claude will receive your instructions, as well as some details about the task. [Click here to see the full prompt we will send to Claude.](#)

Next

FIGURE 6.9: Part I, Task 2: Writing Instructions for an AI

Write Your Instructions for Claude

Please spend at least 1 minute writing your instructions. Time remaining: 59 seconds

Note: Good instructions will provide clear, flexible guidance that can be used across different situations. Bad instructions are too specific to a particular scenario and don't adapt to changing conditions.

You might find it helpful to reference your choices in Task 1 when writing your instructions. [Click here to see your previous choices.](#)

Write your instructions for Claude:

Write clear instructions that describe your preferences for choosing between lotteries. Please write your prompt carefully, as Claude will use these instructions to make choices on your behalf in **new questions**.

While writing your instructions, we ask you to **not browse the internet** and **not consult with others**. We are genuinely interested in your opinions and preferences.

Next

FIGURE 6.10: Write Your Instructions for Claude

6.5 Part I, Task 3: Delegation Choice

Part I, Task 3: Delegation Choice

In this task, Claude will again use the information you provided to make choices on your behalf between **13 new pairs of lotteries (Questions 14–26)**. This time, you will choose what information to send to Claude.

Your Delegation Choice

You can choose **exactly one** (not both) of the following two pieces of information to send to Claude. Once you choose one option, the other option will not be used by Claude.

- **Written instructions (prompt):** If you choose this option, Claude will use the instructions you wrote in Task 2 to infer your preferences and make choices on your behalf for Questions 14–26.
- or
- **Raw choice data:** If you choose this option, Claude will observe your lottery choices from Questions 1–13 in Task 1 and use them to infer your preferences and make choices on your behalf for Questions 14–26.

Next: You will review the information available and make your delegation choice.

Next

FIGURE 6.11: Part I, Task 3: Delegation Choice

Part I, Task 3: Delegation Choice

What Would You Like to Send to the AI?

Review your information above, then click on your choice below:

Your Written Instructions:

"[SUBJECT INSTRUCTIONS HERE]"

Your Task 1 Choice Data:

Show Details

13 lottery choices recorded

Click on your choice:

Written Instructions Only

Send only your written instructions to the AI

Choice Data Only

Send only your lottery choice data to the AI

Next

FIGURE 6.12: Delegation Choice Screen

6.6 Part II: Your Lottery Choices

Part II: Your Lottery Choices

This is the second part of the study, and it contains one task.

In this task, you will make your own choices between **13 new pairs of lotteries**, which are **different from those you saw in Part I**. For each choice, select the lottery you prefer.

Please think about each choice carefully, as one of your choices may be selected to determine your bonus payment.

Next

FIGURE 6.13: Part II: Your Lottery Choices

Question 14 of 26

Click on the lottery you prefer:

Lottery A

Probability	Outcome
95%	\$3.00
5%	\$0.00

Lottery B

Probability	Outcome
91%	\$4.00
9%	\$0.00

Click to compare statistics:

Average Payment

Payment Variability

Minimum Payment

Maximum Payment

Chance of Max

Next

FIGURE 6.14: Lottery Choice Screen (Part II)

Can you spot the animal camouflaged in the image below?

Please click on the photo where you think the animal is located

This task does not affect your payment and is just for fun.



This photo was originally published by the Reader's Digest on August 22, 2019 (<https://www.rd.com/list/camouflaged-animal-photos/>)

Next

FIGURE 6.15: Attention Break

6.7 Part III: Exit Survey

Part III Instructions

This is the third part of the study. In this part, you will answer some short questions that are unrelated and different from all previous questions.

This part is short and should take you about 5-10 minutes to complete.

There are **no right or wrong answers**. We are interested in studying **your preferences**.

In addition to the potential bonus from Parts I and II, you may earn an **extra bonus** in this part. We will explain later how this is determined.

Next

FIGURE 6.16: Part III Instructions

Part III: Your Decision-Making Reasoning

Your Reasoning

Earlier in this experiment, you chose to send **written instructions (prompt)** to the AI to help it make decisions on your behalf.

Why did you make this choice? Please explain your reasoning:

Predict AI Performance

Bonus opportunity: You can earn \$0.25 for each correct prediction below (up to \$0.50 total). This bonus is in addition to your other earnings.

AI with your written instructions (prompt):

If the AI were to answer the same questions (Questions 14–26) on your behalf based on your written instructions, how many of them do you think the AI would have answered the same way as you did?

-- Select a number -- ▾

AI with your lottery choice data:

If the AI were to answer the same questions (Questions 14–26) on your behalf based only on your lottery choices from Questions 1–13, how many of them do you think the AI would have answered the same way as you did?

-- Select a number -- ▾

Next

FIGURE 6.17: Part III: Your Decision-Making Reasoning

Part III: Pattern Matching Test

This part of the study has **6 graphical puzzles**.

Each puzzle you solve correctly pays **\$0.20**.

If you answer all 6 puzzles correctly, you can earn up to **\$1.20**.

For each puzzle, you will see a pattern with a missing piece. Select the answer that best completes the pattern.

Next

FIGURE 6.18: Pattern Matching Test Introduction

Graphical Puzzle 1

Which one comes next?

A 	B 	C 	I don't know
D 	E 	F 	None of these

Next

FIGURE 6.19: Pattern Matching Test: Example Puzzle

Part III: Pattern Matching Test

How many of the 6 previous puzzles do you think you answered correctly?

If that number coincides with the actual number of puzzles you solved correctly, you will receive 25 cents. Otherwise, you will receive 0 cents for this question.

Next

FIGURE 6.20: Pattern Matching Test: Self-Assessment

Part III: Pattern Matching Test

We ran a similar experiment on Prolific and have data from 100 random Prolific participants. How many of these 100 correctly solved more puzzles than you?

Please enter a number between 0 and 100:

Next

FIGURE 6.21: Pattern Matching Test: Relative Performance

Part III: Investment Decision (1 of 2)

You are endowed with **100 tokens** (worth **\$0.50**) and asked to choose the portion of this amount (between 0 and 100 tokens, inclusive) that you wish to invest in a risky option. Tokens not invested are yours to keep.

If the investment is successful: You receive **2.5x** the amount you invested.
If the investment is unsuccessful: You lose the amount invested.

To determine if the investment is successful or not, we will flip a fair coin:

Heads (50%): Success **Tails (50%): Failure**

The coin flip will be performed after you complete the study, and your bonus will be determined accordingly.

How many tokens do you wish to invest?

0 (keep all) 100 (invest all)

I wish to invest this many tokens:

0

Your potential outcomes:

If successful: You will have **100 tokens** (\$0.50)
If unsuccessful: You will have **100 tokens** (\$0.50)

Next

FIGURE 6.22: Investment Decision 1 of 2

Part III: Investment Decision (2 of 2)

You are endowed with **100 tokens** (worth **\$0.50**) and asked to choose the portion of this amount (between 0 and 100 tokens, inclusive) that you wish to invest in a risky option. Tokens not invested are yours to keep.

If the investment is successful: You receive **2.5x** the amount you invested.
If the investment is unsuccessful: You lose the amount invested.

To determine if the investment is successful or not, we will roll a **four-faced die** with faces marked A, B, C, D:

First Roll:

A = Investment is **successful**

D = Investment is **unsuccessful**

B or **C** = Roll the die again

Second Roll (if B or C on first roll):

A or **B** = Investment is **successful**

C or **D** = Investment is **unsuccessful**

The die rolls will be performed after you complete the study, and your bonus will be determined accordingly.

How many tokens do you wish to invest?

0 (keep all) 100 (invest all)

I wish to invest this many tokens:

0

Your potential outcomes:

If successful:

You will have **100 tokens** (\$0.50)

If unsuccessful:

You will have **100 tokens** (\$0.50)

Next

FIGURE 6.23: Investment Decision 2 of 2

Part III: Background Survey

Please answer the following questions about yourself. Your responses will help us understand how different backgrounds relate to decision-making preferences.

AI Experience

How comfortable are you with using AI tools (e.g., ChatGPT, Claude, Copilot)?

Not at all comfortable	Neutral				Very comfortable	
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7

AI Usage

How often do you use AI tools (e.g., ChatGPT, Claude, Copilot) in your daily life or work?

- ☐ Never
- ☐ Rarely (a few times a year)
- ☐ Monthly
- ☐ Weekly
- ☐ Daily

Writing

How comfortable are you with writing instructions or explanations for others?

Not at all comfortable	Neutral				Very comfortable	
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7

Professional Background

Do you have experience managing or supervising other people?

- ☐ No management experience
- ☐ Some experience (1-2 years)
- ☐ Moderate experience (3-5 years)
- ☐ Extensive experience (6+ years)

Education

What is the highest level of education you have completed?

- ☐ High school or equivalent
- ☐ Some college, no degree
- ☐ Associate's degree
- ☐ Bachelor's degree
- ☐ Master's degree
- ☐ Doctorate or professional degree

Personal Traits

How would you rate your level of impatience in general?

Very patient	Neutral				Very impatient	
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7

Next

FIGURE 6.24: Background Survey

Part III: Personality Survey

Instructions: Here are a number of personality traits that may or may not apply to you. Please indicate the extent to which you agree or disagree with each statement. You should rate the extent to which the *pair* of traits applies to you, even if one characteristic applies more strongly than the other.

1. I see myself as extraverted, enthusiastic.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

2. I see myself as critical, quarrelsome.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

3. I see myself as dependable, self-disciplined.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

4. I see myself as anxious, easily upset.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

5. I see myself as open to new experiences, complex.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

6. I see myself as reserved, quiet.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

7. I see myself as sympathetic, warm.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

8. I see myself as disorganized, careless.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

9. I see myself as calm, emotionally stable.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

10. I see myself as conventional, uncreative.

Disagree strongly

Neutral

Agree strongly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

Next

Based on the Ten-Item Personality Inventory (TIPI) by Gosling, Rentfrow, & Swann (2003).

FIGURE 6.25: Personality Survey

Your Results

Experiment Complete

Thank you for completing this study. The payments described below will be sent through Prolific.

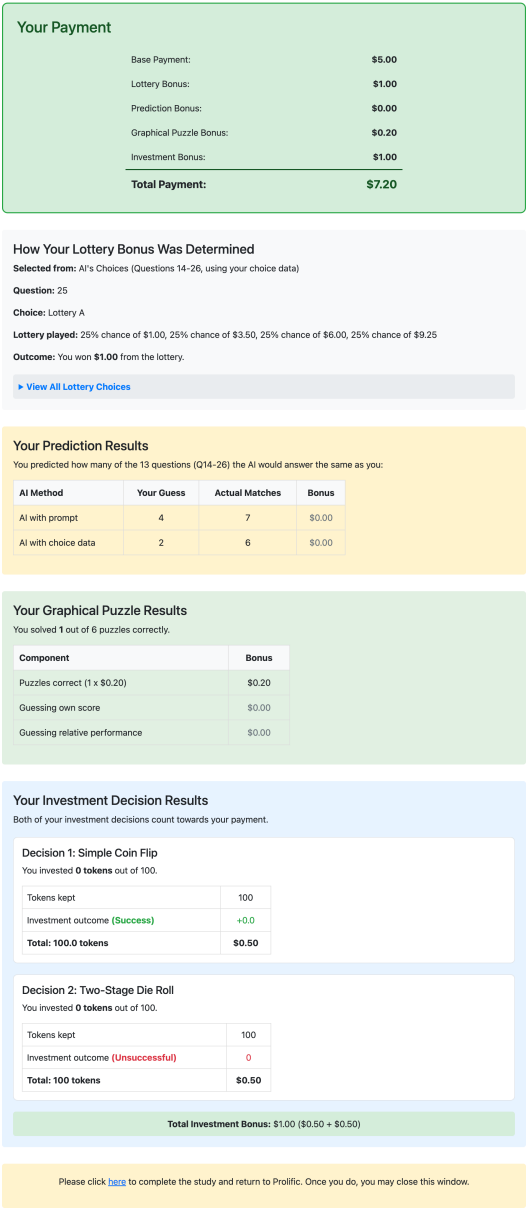


FIGURE 6.26: Payment Summary

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